

Homework1

2026-08-21

```
#options(tinytex.verbose = TRUE)
```

Question 1

Suppose a lab instrument loses 12 of its value each year. Its current value is 1800. Compute its value after 4 years

```
V = 1800*(1-0.12)^4
```

Question 3

Create a vector of temperatures using : 72, 75, 71, 68, 70, 74, 77, 73. Then use indexing to extract: 1. the first temperature, 2. the last three temperatures, 3. all temperatures except the fourth, 4. all temperatures at least 74.

```
## [1] 72
```

```
## [1] 74 77 73
```

```
## [1] 72 75 71 70 74 77 73
```

```
## [1] FALSE TRUE FALSE FALSE FALSE TRUE TRUE FALSE
```

Question 4

Let $x = 2, 4, 6, 8, 10$.

Create this vector in R. Then compute: x^2 , $x + 5$, $\sqrt{x}$, a data frame with columns x , x^2 , $x + 5$, $\sqrt{x}$.

```
x = c(2,4,6,8,10)
# x squared
x_squared = x^2
# x plus five
x_plus_five <- x + 5;
# square root of x
sqrt_x = sqrt(x)
```